

Risk Engine Model Evaluation

Five-session realized-volatility forecasting

Decision. Keep **HAR-X + News** as the formal Risk Engine. Price-only XGBoost is the strongest point-forecast challenger, while XGBoost with a constrained news-attention residual is the best news-aware research challenger. Neither XGBoost variant is ready to replace the formal model.

1 Key findings

- **XGBoost Gamma has the lowest out-of-fold QLIKE: 0.6571**, a **10.31%** improvement over HAR-X ($p = 0.0010$).
- **HAR-X + News is the best production compromise:** QLIKE improves by **5.24%**, the news increment is significant ($p = 0.0011$), and the realized-to-forecast calibration ratio is **0.981**.
- **Residual MLP does not establish a reliable advantage:** average QLIKE improves by 4.64%, but the date-aggregated DM test is not significant ($p = 0.915$).
- **News can improve XGBoost only under a constrained design:** an attention-based Gamma residual reduces XGBoost QLIKE by **1.30%**, but worsens high-volatility QLIKE by **4.17%**.
- **FinBERT sentiment does not improve the XGBoost residual model** in the current panel. Its measurable XGBoost news contribution comes from attention, recency, and causal coverage states.

2 Evaluation protocol

Target	Future five-session daily realized volatility; no price-direction label is used.
Universe	21 FNSPID research symbols; 27,027 stock-date outer-fold forecasts.
Training history	9 April 2013 to 28 December 2023.
Outer evaluation	Annual expanding walk-forward tests for 2018–2023.
Leakage control	Five-session embargo at each outer boundary; model, feature, and calibration decisions use prior data only.
Weighting	Stock-equal loss weighting so high-news-volume symbols cannot dominate.
Primary metric	QLIKE on realized variance; lower is better.
Supporting evidence	Log-RMSE, log- R^2 , calibration ratio, high-volatility QLIKE, per-stock gains, Newey-West DM tests, and moving-block bootstrap intervals.

3 Models

3.1 HAR-X

HAR-X is an interpretable pooled log-volatility model. The formal price specification uses six causal features:

$$\log \hat{\sigma}_{t,5}^P = \beta_0 + \beta_1 \log RV_{t,5} + \beta_2 \log RV_{t,22} + \beta_3 \log RV_{t,66} + \beta_4 \log PARK_{t,5} + \beta_5 \log PARK_{t,22} + \beta_6 \log |r_t|.$$

The fitted log forecast is exponentiated and corrected with a smearing factor. HAR-X provides a smooth and interpretable reference, but misses more of the high-volatility regime than the nonlinear challengers.

3.2 HAR-X + News

The formal five-session model combines the HAR-X price forecast with a regularized Gamma variance-ratio term:

$$\hat{\sigma}_{t,5} = \hat{\sigma}_{t,5}^P \times \exp \left[\frac{1}{2} \gamma \frac{\log(1 + N_t)}{s_N} \right].$$

Here N_t is the causal news count and s_N is the training-period scale. A zero-news observation is retained with $N_t = 0$; it is not dropped or replaced by a sentiment-neutral article. The current deployable news term uses `log_count`, selected in all six outer folds.

3.3 XGBoost Gamma

The XGBoost challenger uses a Gamma objective to predict positive realized variance from 34 price features. The inputs include HAR features, OHLC range estimators, semivariances, jump and extreme-return measures, volume, liquidity, SPY market risk, and cross-sectional risk states. For each outer year, shallow-tree hyperparameters are chosen from the original 96-configuration inner search, with early stopping and an internally selected HAR blend.

3.4 Residual MLP

The neural candidate uses 68 inputs: 34 price features and 34 deployable news/state features. Three regularized architectures are considered inside each outer training window:

$$64-32, \quad 128-64, \quad 128-64-32.$$

The networks use LayerNorm, GELU activations, dropout, AdamW, and a bounded log-volatility residual around a fold-local HAR forecast. The candidate has 6,721–19,649 trainable parameters depending on the selected architecture.

4 Core out-of-fold results

Table 1: Five-session point-forecast performance, 2018–2023 OOF. Lower QLIKE and log-RMSE are better.

Model	QLIKE	Gain	Log-RMSE	Log- R^2	Calib.	High-vol	DM p
XGBoost Gamma	0.6571	10.31%	0.5685	0.291	0.892	1.3539	0.0010
HAR-X + News	0.6943	5.24%	0.5400	0.360	0.981	1.6808	0.0011
Residual MLP	0.6987	4.64%	0.5567	0.320	0.917	1.5654	0.9151
HAR-X	0.7327	Reference	0.5192	0.408	1.057	1.9986	–

Interpretation. XGBoost is the strongest model under the primary QLIKE objective and produces the lowest high-volatility loss. HAR-X has better ordinary log-error metrics, which reflects smoother performance in normal regimes, but it underperforms on variance-ratio loss. HAR-X + News gives up some point-forecast performance relative to XGBoost in exchange for a near-unit calibration ratio, validated news contribution, simple serialization, and stable zero-news behavior.

The MLP improves in five of six annual folds and 81.0% of stocks, but the daily loss-difference series is unstable enough that its aggregate DM p -value remains 0.915. It is not a promotion candidate.

5 Validated news contribution

5.1 Formal HAR-X + News

The formal news model passes the historical accuracy, breadth, significance, and tail-calibration gates. Its only unresolved evidence requirement is a mature live-RSS evaluation period.

5.2 Can news improve XGBoost?

Two architectures were tested without removing zero-news samples.

Table 2: News increment and tail-risk validation for the formal model.

Metric	Result
Stock-equal QLIKE gain vs HAR-X	5.24%
95% moving-block bootstrap interval	+2.38% to +8.83%
DM test	$p = 0.001087$
Positive outer years	5/6
Positive stocks	85.7%
Median stock gain	7.51%
High-volatility QLIKE gain	13.56%
VaR-95 breach rate	4.32%
95% risk-band coverage	96.34%
Expected-shortfall ratio	0.983

Direct concatenation failed. Appending news fields to the 34 price inputs increased QLIKE from 0.6571 to 0.6855, a **4.32% deterioration**. News-active rows deteriorated by 2.73%, high-volatility rows by 7.93%, and news features represented only 1.67% of final-tree total gain.

A constrained residual layer found a smaller positive signal. The price-only XGBoost forecast was held fixed and news was used only to model its remaining variance ratio:

$$\hat{\sigma}_{t,5} = \hat{\sigma}_{t,5}^{XGB} \times \text{GammaRatio}(\text{news}_t).$$

Table 3: News residual families on top of price-only XGBoost.

News family	QLIKE	Gain	DM p	Bootstrap 95%	Years	Stocks	High-vol
Attention	0.6486	+1.30%	0.00137	+0.51% to +2.67%	4/6	81.0%	-4.17%
Log count	0.6531	+0.61%	0.00027	+0.27% to +1.39%	5/6	71.4%	-2.58%
All deployable	0.6518	+0.81%	0.05235	-0.10% to +2.44%	4/6	71.4%	-4.58%
Attention + FinBERT	0.6627	-0.84%	0.64580	-3.12% to +1.89%	4/6	57.1%	-8.12%

The attention family contains article frequency, short-window counts, time since the latest article, and causal coverage/silence states. Its average gain remains significant after a conservative four-family multiple-test correction, but it fails the high-volatility non-degradation gate. FinBERT polarity and distribution fields do not add value to this XGBoost residual specification.

6 External generalization and production status

The independent 2024–2026 price-backbone test contains 40 symbols and 25,400 mature five-session observations.

Table 4: HAR-X price-backbone external validation against trailing-22-day volatility.

Scope	Symbols	HAR QLIKE	Baseline QLIKE	Gain	DM p
All external panel	40	0.5829	0.6354	8.26%	0.00345
Original research symbols	21	0.6661	0.7195	7.42%	0.02742
Unseen symbols	19	0.4910	0.5425	9.48%	0.00626
Current six-symbol portfolio	6	0.6181	0.6262	1.30%	0.82950

Across all 40 symbols, the VaR-95 breach rate is 4.60%, the 95% risk-band coverage is 95.37%, and the expected-shortfall ratio is 0.914. The unseen scope is mostly broad and sector ETFs, so it is evidence of cross-symbol transfer rather than a complete unseen-single-stock test.

The live news contract remains the final production gap. FNSPID validates the historical news increment, but the system still needs at least 60 mature five-session RSS forecasts to confirm that live source intensity and coverage transfer correctly.

7 Final model decision

1. **Formal output: HAR-X + News.** It is the only candidate that combines a statistically significant news contribution with acceptable VaR, risk-band, expected-shortfall, and high-volatility behavior.
2. **Primary challenger: price-only XGBoost Gamma.** It has the best OOF QLIKE, but needs external unseen-stock validation, component-specific tail calibration, and a news specification that does not degrade high-volatility performance.
3. **News-aware XGBoost research ceiling: attention residual.** The +1.30% average gain is real, but the -4.17% high-volatility result blocks promotion.
4. **Residual MLP: research only.** More cross-sectional data and a longer live-news history are required before neural complexity can be justified.

Reproducibility artifacts

- Core comparison: `reports/risk_engine_presentation/model_comparison.csv`
- Annual OOF results: `reports/risk_engine_presentation/yearly_qlike.csv`
- XGBoost news-family comparison: `reports/risk_engine_xgb_news/overlay_family_comparison.csv`
- Formal promotion gates: `reports/risk_engine_optimization/promotion_gates.json`
- Current external validation: `data/processed/current_risk_test/current_validation_summary.csv`
- Formal checkpoint: `data/processed/risk_model.json`